

Integrating Communications and Surveillance Technologies for effective community policing in Nigeria

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ABSTRACT

My research investigates how technologically enhanced community policing (CP) approach can curb crime and insecurity in Nigeria, by efficiently integrating communications and surveillance systems. In order to achieve this, I have conducted a narrative literature review to understand the current state of community policing in Nigeria, and two systematic literature reviews; firstly to comprehensively understand trust and relationship issues between Nigerian police and the citizens, and secondly to identify technologies used in community policing since inception and how each has fared, as to gain better design insight and direction. Also, I have conducted a survey and interview with 1200 and 18 Nigerian citizens respectively, and a separate interview with 6 Nigerian police commissioners representing the 6 geopolitical zones of the country, through which I gathered relevant data as regards the use of technologies in Nigeria to reduce crime and insecurity. The outcome of the survey and interviews informed a design of a novel body-worn device, integrated with a mobile application for real-time information dissemination. My research findings will produce a tool that will improve community policing in Nigeria and lower insecurity and crime rates therein.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Empirical studies in interaction design;

KEYWORDS

Community Policing, Surveillance, Communications, Technologies, Crime

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1 INTRODUCTION

The concept of community policing (CP) has gained significant attention worldwide as a model that promotes collaboration between law enforcement agencies and the communities they serve [5]. In the Nigerian context, community policing is an essential strategy for addressing the diverse and evolving security concerns that confront the nation. However, the effective implementation of community policing in Nigeria has been hindered by lack of modern tools and technologies [16]. This dissertation explores the vital intersection of communications and surveillance technologies and their role in transforming the landscape of community policing in Nigeria. Moreover, the study will assess the potential benefits and risks associated with the adoption of such technologies, including issues related to privacy, security, and public trust. Through a comprehensive analysis of existing CP strategies in Nigeria and beyond, this dissertation will provide insights and recommendations for policymakers, law enforcement agencies, and other stakeholders seeking to enhance the effectiveness of community policing in Nigeria. It is my hope that this study will contribute to the development of a more secure and connected future for Nigerian communities, where safety and trust are paramount, and where technology plays a pivotal role in achieving these objectives.

1.1 PROBLEM STATEMENT

The problem at hand is twofold:

Firstly, Nigerian law enforcement agencies currently face limitations in their ability to harness the full potential of communications and surveillance technologies [3, 16]. The existing infrastructure often lags behind the technological advancements seen in other parts of the world [7]. This lack of integration and modernization leads to a disconnect between law enforcement agencies and the communities they serve [19], resulting in slower response times to crime, suboptimal information sharing between the police and citizens, and an inability to harness real-time data for effective policing and curbing insecurity [10, 20]. Secondly, the principles of community policing, such as building trust, maintaining transparency, and fostering community partnerships, has been challenging to implement in Nigeria [8]. Consequently, the goal of addressing local crime and safety issues through cooperation and engagement is often undermined by the absence of up-to-date communication and surveillance systems. My dissertation aims to investigate the extent of these challenges, analyze the consequences of the limited integration of communication and surveillance technologies in the context of community policing in Nigeria and design a more efficient system for overcoming them. Furthermore, my research seeks to contribute to the development of a more effective and responsive community policing model that can better serve the security needs

of Nigerian communities. The next section will detail the gap in literature and the relevance of this PhD research, and the following section will summarily look at related works. The subsequent section will detail the research questions and hypothesis, while the last section will talk about the methodology of the research.

1.2 RELEVANCY

While existing studies have explored the use of SMS, phone calls and social media technologies to foster policing practices in Nigeria [6, 13, 18], nothing is known about the integration of communications and surveillance technologies in addressing the complex security threats of terrorism, communal clashes, kidnapping, cybercrime, and organized crime. These modern technologies include real-time data sharing, improved communication channels, and enhanced surveillance capabilities, which are critical for an effective response to crime and emergencies [9, 11, 20]. Hence, technologically enhanced community policing is a critical area of study with the potential to significantly improve the safety and security of Nigerian communities.

2 RELATED WORKS

There is paucity of study regarding technological application in community policing in Nigeria. However, some authors who have looked into this have done well, though not detailed. For instance, a research by Onyeozili et al. [16] posit that community policing has been a successful tactic for resolving security and criminal issues in Nigeria. Their study emphasises the concepts and historical background of community policing in Nigeria, with a focus on how it may foster collaboration and trust between law enforcement and communities. I have also looked at other studies with six relevant subheadings as shown below.

2.1 Challenges in Community Policing Implementation

Some academics have highlighted the difficulties Nigerian community policing faces [2, 15]. According to Taiwo [18], problems include inadequate community involvement, a misalignment of policy and practise, and a lack of resources and training. These difficulties highlight the need for innovation, which includes fusing technologies for monitoring and communication.

2.2 Technology Integration in Community Policing

Research from other countries demonstrates the positive impact of technology integration in community policing. A study conducted by Lewis and Lewis [10] in the United States reveals that predictive policing models, enabled by technology, can improve the allocation of resources and reduce crime rates. These findings emphasize the potential for technology to enhance community policing efforts in Nigeria.

2.3 Surveillance Technologies and Crime Prevention

There is ample evidence supporting the use of surveillance technologies to reduce crime. For example, a research done by Calacci

et al.[4] highlights how surveillance cameras can help law enforcement with their investigations and discourage criminal activity. In Nigeria, comparable technology can be used to raise community safety.

2.4 Privacy and Ethical Concerns

As technology integration progresses, ethical and privacy issues surface. The ethical issues surrounding the use of surveillance technology are covered in a research by Sheng et al. [17], along with the necessity of explicit legislation to protect citizens' privacy. This emphasises how crucial it is to take these issues into account when integrating surveillance technologies into Nigerian community police.

2.5 International Best Practices

Nigeria can benefit from other countries' experiences. For instance, the use of technology to improve community participation, trust, and crime prevention is emphasised in the United Kingdom's approach to community policing, as described in studies by Zilka et al. [21].

2.6 Public Perception and Trust

The public's confidence and trust are essential for the success of community policing. Research by Alikhademi et al.[1] and Zilka et al. [21] emphasise the importance of transparency and communication in fostering public trust. Effective use of technology will facilitate the building of confidence and trust in the Nigerian context.

3 RESEARCH QUESTIONS

The main research question in my doctoral research is: *How can the integration of communication and surveillance technologies contribute to enhancing the effectiveness of community policing in Nigeria?* This question gives rise to three others.

- What are the correlation between the level of technological infrastructure implementation and the effectiveness of community policing in Nigeria?
- What are the correlation between public trust in surveillance technologies and the perceived effectiveness of community policing efforts in Nigeria?
- To what extent do ethical and legal considerations impact the integration of communication and surveillance technologies for community policing?
- How does the relationship between ethical and legal considerations and integration of communication and surveillance technologies impact overall community satisfaction with policing efforts in Nigeria?

4 HYPOTHESES

The main hypotheses is *The integration of communication and surveillance technologies does not contribute to enhancing the effectiveness of community policing in Nigeria* Other hypotheses are:

- There is no significant correlation between the level of technological infrastructure implementation and the effectiveness of community policing in Nigeria.

- There is no significant correlation between public trust in surveillance technologies and the perceived effectiveness of community policing efforts in Nigeria.
- Ethical and legal considerations do not significantly impact the integration of communication and surveillance technologies for community policing in Nigeria.
- The relationship between ethical, legal considerations and integration of communication and surveillance technologies does not impact overall community satisfaction with policing efforts in Nigeria.

5 METHODOLOGY

5.1 Research Design

I employ a mixed-methods research design [12] to comprehensively investigate how the integration of communication and surveillance technologies contribute to enhancing the effectiveness of community policing in Nigeria.

5.2 Data Collection

- **First Study (Narrative literature review):** The goal of the first study is to broadly understand the present practice of community policing in Nigeria as to identify gaps and trends in the literature. I conducted a narrative literature review which led to qualitative synthesis of existing research on community policing and why insecurity is rising amidst the strategy in Nigeria. I found out that trust was a major challenge to the strategy in Nigeria. However, the term ‘trust’ was used ambiguously.
- **Second Study (Systematic literature review):** To understand the trust being talked about in community policing, I conducted a systematic literature review based on trust in community policing. This reduced the ambiguity and gave rise to a manuscript which I presented in IEEE International symposium on Technology and society in October 2023 [14].
- **Third Study (Survey):** The third study has the goal of gathering first-hand data from Nigerian citizens and specific information that can inform my research, validate my hypotheses, and help me draw meaningful conclusions and make informed design decisions. I conducted a survey with 1200 Nigerians on technology usage, community engagement, and the impact of technology on policing.
- **Fourth Study (Interview):** The goal of this study is to gather in-depth, qualitative insights from Nigerian citizens on their personal experiences relevant to my research. I conducted in-depth interviews with key stakeholders, including law enforcement officials and community members to gain insights into their perspectives on technology integration, challenges, and potential solutions. This study uncovered unique and valuable insights that was not accessible through the other research methods, and provided rich and nuanced design.
- **Fifth Study (Systematic literature review):** The goal of this study is to identify technologies applied in community policing and how each has fared around the globe as to receive clearer direction. I conducted a systematic literature review which gave rise to a manuscript with the title “An

Empirical Analysis of Technological Applications in Community Policing and Their Performance Assessment Through A Systematic Review”. This paper is still under consideration by a reputable journal. This study provided me with design options and directions which will help towards achieving my research aim.

6 ONGOING WORK:

I have started gathering the required hardware and software tools to design a surveillance and communications body-worn device, which will help answer my research questions.

7 REFLECTION

The major discrepancy in my approach compared to existing works on technologically enhanced community policing research is that I incorporate a more extensive mixed method to understand the situation at hand from both the citizens and police more broadly, and I intend to develop a novel device that integrates surveillance and communication features, which can be adopted and used by all CP stakeholders to contribute in curbing crime and insecurity.

8 EVALUATION

The evaluation of this study will focus on the performance of the system to be developed. Hence there are two focal lines.

8.1 Citizens and the police user research

Since the design of the device is informed by survey and interview of both Nigerian citizens and the police, it will be of relevance to organise a user study and to assess its usability, user satisfaction, and privacy concerns. These participants will be asked to interact with the device, as to provide valuable feedback on its interface, features, and overall user experience.

8.2 Design modification:

At this phase, modification will be done on the device based on what the user research reveals. Their appreciation of the device’s communication capabilities and convenience, their expressions on significant privacy reservations, particularly regarding data security and the potential for unauthorized access will all be taken into considerations. This is to ensure that there is successful adoption and long-term acceptance by Nigerian citizens and the police.

9 CHALLENGES

There is no doctoral research that doesn’t have its challenges. This section enumerates some of the challenges.

- Integrating surveillance and communications technologies involves knowledge from both the realms of policing and technology. Hence, balancing expertise in both fields and getting police participants willing to help in all phases of the data collection have been challenging.
- Secondly, since surveillance technologies raises ethical questions about privacy and civil liberties, addressing these concerns while advancing the technology for policing purposes is a challenge.

- Lastly, implementing advanced technologies in community policing requires financial and human resources, and striking a balance between the need for effective solutions with the available resources is a pressing challenge.
- A future challenge will be the smooth adoption and integration of these tools into existing practices generally.

10 BENEFITS AND CONTRIBUTION

The upcoming doctoral consortium will offer significant benefits to me at this stage of my doctoral program. I will have the chance to present my research, receive constructive feedback, and gain new perspectives on my work, ultimately helping me refine my research direction and methodology. My research has the potential to spark interesting discussions and inspire others in the consortium. Also, I will offer constructive feedback on the works of my peers and exchange contact information for potential collaboration and further discussions. I will make myself available for one-on-one conversations to delve deeper into people's research, provide guidance in methodical challenges and share relevant research papers and HCI resources that have helped me in my research.

11 CONCLUSION

This paper has presented the outline of my PhD research which focuses on integrating surveillance and communication systems to reduce crime and insecurity in Nigeria. This work contributes a comprehensive understanding of the current technological landscape and its real-world implications, offering valuable insights for law enforcement agencies and policymakers to enhance community safety and engagement. It will benefit Human Computer Interaction (HCI) designers and community policing researchers by providing design implications and research directions through which challenges in this context would be overcome, as to reduce crime and insecurity in Nigeria and beyond.

12 LONG-TERM GOAL

My long term goal is to develop methodologies to assess the effectiveness and sustainability of the proposed solutions over time, as to intermittently evaluate the long-term impact of integrated technologies on community policing practices across the globe.

13 RESEARCH SITUATION

I am a second year PhD candidate (fulltime) in the Computer Science Department, at Swansea University, United Kingdom. My projected year of completion is 30th of June, 2025. However, I can apply for extension of the program if I am unable to complete my research at the projected date. This will usually take about six months if granted. I have a prime focus on community policing and technologies for crime reduction and will consider myself a HCI researcher because of my pattern of research and methods I apply.

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